

Katie Li

☎ 650-285-8450 ✉ katieli167@gmail.com 🔗 [linkedin.com/in/katieli167](https://www.linkedin.com/in/katieli167) 👤 katie-li.vercel.app

Education

University of Illinois at Urbana-Champaign

Bachelor of Science in Computer Science (GPA: 4.0/4.0)

Expected Graduation: Dec 2027

Champaign, IL

- **Honors:** 2025 Illinois Engineering Achievement Award | Rothmuller Award | Dean's List
- **Coursework:** Algorithms & Models of Computation, Data Structures, Computer Architecture, Prob & Stats in CS

Technical Skills

Languages: Python, C++, Java, Go, JavaScript/TypeScript, SQL

Frameworks & Tools: React, PostgreSQL, TensorFlow, PyTorch, Pandas, Docker, AWS, Git

Certifications: Machine Learning Specialization (DeepLearning.AI, Stanford)

Experience

Software Engineer Intern

Uber

Jan 2026 – Present

San Francisco, CA

- Implemented an **ETA-based reoffering** feature to dynamically update ride offers as drivers move closer, reducing rider wait time and improving overall completion rate.
- Ran shadow tests and canary experiments to optimize the ETA threshold, extending existing data models and APIs across **two production microservices** (Go, Java) to support ETA extraction.

Software Developer

PrairieLearn

Sep 2025 – Jan 2026

Champaign, IL

- Built interactive modules using **Python**, **HTML**, and **JavaScript** for the University of Illinois' online testing platform.
- Worked with course professors and implemented new grading logic and assessment features across core CS courses (CS 374, 225, etc), used by over **4,000 students** annually.

Machine Learning Researcher

Forward Data Lab, University of Illinois

Aug 2025 – Dec 2025

Champaign, IL

- Engineered an **agentic retrieval-augmented generation (Agentic RAG)** system integrating GPT-4o with **FAISS**-based vector databases to power semantic retrieval of academic research papers and automated literature search.
- Implemented the pre-processing of over **320k+** arXiv papers, generating and indexing **MiniLM vector embeddings**.

Research Assistant

College of IST, Pennsylvania State University

May 2025 – Jul 2025

University Park, PA (Remote)

- Wrote **Python batch processing pipelines** to generate evaluations of instructional videos, utilizing Whisper for audio extraction and GPT-4o for analysis.
- Scaled the system to process **400 hours** of video with 80% evaluator alignment, under submission to WACV 2026.

Web Development Intern

Robotics For All

Jan 2022 – Jul 2022

Cupertino, CA

- Revamped production website features in **React** and **JavaScript**, supporting nonprofit outreach to **3,000 users**.
- Deployed a new donations workflow, integrating **Google Forms** with **backend services** to streamline fundraising.

Projects

Bookrates: Book Recommendation Platform | React, Flask, AWS, Python, JavaScript, CSS

- Built and deployed a **full-stack** website with a **React** frontend and **Flask** backend hosted on **AWS EC2**.
- Developed **REST API** endpoints to **process recommendation queries** and return ranked JSON results, integrating with a **database of 70k+ books** and **1M+ user interactions**.
- Optimized system performance through adding Parquet-based caching layers, reducing query times by **2x**.

Stride: Mobile Fitness App | TypeScript, React Native, PostgreSQL, XCode, TensorFlow

- Built a cross-platform fitness app using **React Native + TypeScript**, with run tracking features and progress graphs.
- Implemented a **PostgreSQL backend** with **Supabase**, supporting user authentication and credential storage.
- Integrated **MoveNet** for real-time inference, enabling instant **AI powered pose detection** and analysis.